

Improvement of gelling properties of lizardfish mince as influenced by microbial transglutaminase and fish freshness.

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The effects of microbial transglutaminase (MTGase) at different levels (0 to 0.8 units/g sample) on the properties of gels from lizardfish (*Saurida undosquamis*) mince set at 25 degrees C for 2 h or 40 degrees C for 30 min prior to heating at 90 degrees C for 20 min were studied. Breaking force and deformation of gels increased with increasing MTGase amount added ($P<0.05$). At the same MTGase level used, gels with the prior setting at 40 degrees C for 30 min showed a higher breaking force compared with those subjected to prior setting at 25 degrees C for 2 h ($P<0.05$). Sodium dodecyl sulfate-polyacrylamide gel electrophoretic study revealed that myosin heavy chain (MHC) underwent polymerization to a higher extent in the presence of MTGase. Regardless of setting condition, microstructure of gel added with MTGase was finer with a smaller void compared with that of gel without MTGase. Therefore, setting temperature affected the property of gels added with MTGase. Gel properties of mince obtained from lizardfish stored in ice for different times (0 to 10 d) with and without MTGase at a level 0.6 units/g were determined. Irrespective of MTGase addition, breaking force and deformation of all gels decreased as the storage time of lizardfish increased ($P<0.05$). The addition of MTGase was able to increase both breaking force and deformation of the resulting gel produced from lizardfish kept in ice for all storage times used. Therefore, both freshness and MTGase addition had the direct impact on gel properties of lizardfish mince.